AR(1) $\quad y_t = c + \phi_1 y_{t-1} + \varepsilon_t$

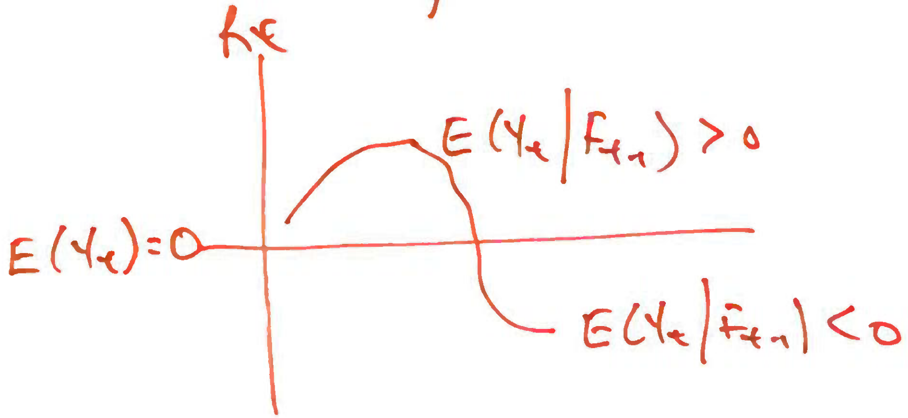
$\uparrow$ Assume e.g. ε_t MDS $\quad E(\varepsilon_t | F_{t-1}) = \cancel{a}\,\sigma^2$

RV. $\quad E(Y_t | F_{t-1}) = c + \phi_1 Y_{t-1}$ $\quad$ SR

$E(Y_t) = \dfrac{c}{1-\phi_1}$ $\quad$ LR

Say $c = 0$

sample mean. $\Rightarrow$



$$\varepsilon_t = u_t \sigma_t$$

$\downarrow$

Assume u_t.

① W8 Slide 42 Assume $E(u_t \mid F_{t-1}) = 0$ $E(u_t^2 \mid F_{t-1}) = 1$

② W8 Slide 46 Stronger Assumption $u_t \sim iid\ N(0,1)$

$\downarrow$ standardised news $\Rightarrow$ W9 Slide 25

is wrong too. $\leftarrow$ $u_t \sim iid\ N(0,1)$

$\frac{\varepsilon_t}{\sigma_t} \sim iid\ N(0,1)$

$\hookrightarrow$ σ_t is known given F_{t-1} !!! $\Rightarrow$ $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$ ARCH(1)

$$\boxed{\varepsilon_t \mid F_{t-1} \sim N(0, \sigma_t^2)}$$

write ① $E(\varepsilon_t \mid F_{t-1}) = 0$ ③ $E(\varepsilon_t^3 \mid F_{t-1}) = 0$

② $V(\varepsilon_t \mid F_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$ Assume.

$\hookrightarrow$ ARCH(1) is wrong.

Big picture : Assume u_t

provides ~~Condina~~ Conditional info on $\varepsilon_t \Rightarrow$ Box

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Given those Assumption, we derive

$E(\varepsilon_t) = 0$ sample mean

$LR \Rightarrow V(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1} =$ sample variance = a number

$E(\varepsilon_t^3) = 0$

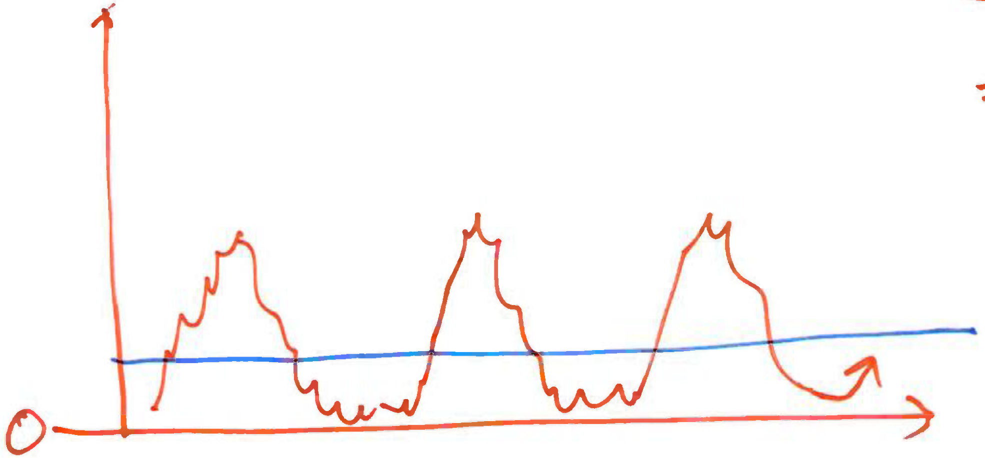
RV

$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$ = SR

$K > 3$ sample info.

$\rightarrow$ fat tail $\rightarrow$ we see this sample w2

$\Rightarrow$ Not Normal.



$V(\varepsilon_t)$ = a number. $\Rightarrow$ LR

① $E(\varepsilon_t) = 0$

$$E(\varepsilon_t) = E\big(\underbrace{E(\varepsilon_t \mid F_{t-1})}_{R.V.}\big)$$

$$= E(0)$$

$$= 0$$

Box

$$\varepsilon_t \mid F_{t-1} \sim N(0, \sigma_t^2)$$

$$E(\varepsilon_t \mid F_{t-1}) = 0$$

$$V(\varepsilon_t \mid F_{t-1}) = \sigma_t^2$$

$$= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$$E(\varepsilon_t^3 \mid F_{t-1}) = 0$$

Assumption.

conditional $\Longrightarrow$ LIE $\Longrightarrow$ unconditional.

$$v(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1} \quad \text{to show} \simeq 2.37\%$$

$t=2 \quad \hat{\varepsilon}_{t-1} = \hat{\varepsilon}_1$

$$V(\varepsilon_t) = E\left(\varepsilon_t - E(\varepsilon_t)\right)^2$$

$$= E\left(\varepsilon_t^2\right)$$

$$= E\left(\underbrace{E(\varepsilon_t^2 \mid F_{t-1})}\right)$$

$$= E\left(\sigma_t^2\right)$$

$$= E\left(\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2\right) \rightarrow \text{GARCH}(1,1)$$

$$= \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2) + \beta_1 E(\sigma_{t-1}^2)$$

$$= \alpha_0 + \alpha_1 E(\varepsilon_t^2) + \beta_1 E(\sigma_t^2)$$

$$= \alpha_0 + \alpha_1 v(\varepsilon_t) + \beta_1 v(\varepsilon_t)$$

$$v(\varepsilon_t \mid F_{t-1}) = \sigma_t^2$$

$$E(\varepsilon_t^2 \mid F_{t-1}) = \sigma_t^2$$

$$E(\varepsilon_t \mid F_{t-1}) = 0$$

$$\boxed{\underset{\text{solve}}{\Rightarrow} \quad v(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1}}$$